

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	45.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Lower abdominal discomfort;Urinary urgency

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	8920.0	cells/ μ L	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.0	mg/dL	3.5 - 7.2
Blood Urea	24.0	mg/dL	7 - 20
Cue Pus Cells	11.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clarithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 480 mg IV over 30 minutes daily for a 80 kg patient)

Precautions: Monitor serum CPK weekly for myopathy; ensure renal function remains stable (dose adjustment not required for CrCl >30 mL/min); avoid use in patients with known hypersensitivity to daptomycin; not indicated for pulmonary infections; counsel patient on possible muscle pain and report promptly.

Rationale: Enterococcus faecium is predicted to be resistant to many agents and may exhibit vancomycin resistance (VRE). Daptomycin is the only antibiotic listed as sensitive and is active against VRE, making it the appropriate choice for this recurrent urethritis.

Antibiotic Drug History

Clinical Summary

Infection Summary

- **Type / Site:** Recurrent urethritis (lower urinary tract) in a 45-year-old female.
- **Predicted Pathogen:** *Enterococcus faecium*, a gram-positive cocci with a high likelihood of vancomycin resistance (VRE).

Antibiotic Susceptibility

- **Resistant:** Clarithromycin (predicted).
- **Sensitive:** Daptomycin (only agent predicted to retain activity).

Recommended Therapy

- **Drug:** Daptomycin 6 mg/kg IV once daily (e.g., 480 mg IV over 30 min for an 80-kg patient).

- **Rationale:** *E. faecium* is often multidrug-resistant and may be VRE; daptomycin is active against VRE and is the sole predicted sensitive option for this infection.

Key Precautions & Monitoring

- Check baseline and weekly serum creatine phosphokinase (CPK); discontinue if CPK rises $>5\times$ ULN or if patient develops myopathy.
- Renal function is stable ($Cr \approx 0.7$ mg/dL); dose adjustment not required for $CrCl > 30$ mL/min.
- Avoid use in patients with known daptomycin hypersensitivity.
- Not indicated for pulmonary infections; ensure patient understands to report new muscle pain promptly.

Clinical Action

Initiate daptomycin as above, monitor CPK and renal parameters, and reassess after 48-72 h for clinical response. Adjust therapy if culture or susceptibility data become available.